Smart Delivery Operations Analytics System

A Real-World SQL Project for Junior Data Analysts

Project Type	Industry-Simulation SQL Analytics Project
Domain	Food Delivery / E-Commerce Operations
Role	Junior Data Analyst — Business Intelligence Team
Tools Required	MySQL / PostgreSQL, SQL Workbench or DBeaver
Difficulty	Beginner to Intermediate
Dataset Size	10,000 Orders | 1,000 Customers | 100 Restaurants | 200 Partners

#SQL #DataAnalysis #FoodTech #BusinessIntelligence #DataCleaning #WindowFunctions #CTEs #RealWorldProject

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1. Project Overview

In today's fast-paced digital economy, food delivery platforms such as Swiggy and Zomato process tens of thousands of orders every single day. Every order generates a rich trail of structured data — customer profiles, restaurant performance metrics, delivery timings, payment transactions, and post-delivery ratings.

While the volume of data collected is enormous, data alone does not create value. Organisations hire **Data Analysts and SQL Developers** specifically to transform this raw, unprocessed data into clear, actionable insights that guide leadership in making better operational and strategic decisions.

In this project, you will step into the role of a **Junior Data Analyst** working for a fictional food delivery company. You will use SQL as your primary analytical tool to explore real-world business problems, detect patterns in operational data, and build a complete analytics system from the ground up.

10,000+ Orders	1,000 Customers	100 Restaurants	200 Delivery Partners	7,000 Ratings

2. Business Scenario

The company has been operating successfully for three years and has expanded its services across multiple cities in India. However, management has recently begun noticing warning signs that threaten customer satisfaction and profitability.

Delayed Deliveries	Customer complaints about late deliveries have surged, directly impacting satisfaction scores and repeat orders.
Rising Cancellations	Order cancellation rates have climbed over the past six months, causing significant revenue leakage.
Inconsistent Revenue	Revenue growth varies widely across cities and restaurant categories with no clear explanation.
Customer Churn	Several loyal customers have gone inactive, and new customer acquisition costs are increasing.
Partner Performance	Some delivery partners are consistently underperforming, affecting delivery speed and customer ratings.

Management has collected all operational data but **lacks the analytical capability** to understand the root causes. The Analytics Team — your team — has been tasked with investigating this data and producing reports that will directly influence business strategy.

3. Project Goals

The primary objective of this project is to build a **complete, end-to-end SQL-based analytics system**. This system must be capable of answering real business questions across five operational dimensions:

Dimension	Focus Area
Customer Behaviour	Understand ordering patterns, frequency, and churn signals.
Restaurant Performance	Identify top and underperforming restaurant partners.
Delivery Efficiency	Measure and improve delivery speed by partner and city.
Revenue Generation	Track revenue trends by city, cuisine, and payment method.
Business Growth	Identify expansion opportunities and seasonal patterns.

To achieve these goals, students will:

#	Item
1	Store and manage structured business data using proper relational design.
2	Establish relationships between multiple tables using primary and foreign keys.
3	Clean and validate raw imported data to ensure analytical accuracy.
4	Write SQL queries ranging from basic SELECT statements to advanced window functions.
5	Generate analytical reports that answer at least 40 business questions.
6	Identify trends, patterns, and root causes from real operational data.
7	Provide data-backed business recommendations to management.
8	Prepare a clean dataset for future Data Science and Machine Learning projects.

4. Student Role & Responsibilities

You are working as a **Junior Data Analyst** in the Business Intelligence Team. Your reporting manager has provided you with six operational datasets and a set of critical business questions that need answers.

Step 1 **Understand the Data**	Study the structure of each table. Understand the column definitions, data types, and how each table relates to the others. Before writing a single query, you must know what the data represents.
Step 2 **Design the Database**	Create an Entity-Relationship (ER) Diagram. Define primary keys, foreign keys, and the cardinality of each relationship. A properly designed schema is the foundation of every reliable analytics system.

Step 3 **Load Data**	Import all six CSV files into MySQL. Validate record counts after each import to confirm the data loaded correctly and completely.
Step 4 **Clean the Data**	Raw data is never perfect. You will identify and fix NULL values, duplicate records, incorrect city names, out-of-range ratings, and missing delivery timestamps. Document every cleaning action you take.
Step 5 **Analyse Business Performance**	Write SQL queries to answer management's questions across delivery, cancellations, revenue, customers, and partner performance. Use the full range of SQL — from basic aggregations to CTEs, subqueries, and window functions.
Step 6 **Generate Insights**	Do not just produce query output. Explain what you found, why it matters, and what management should do about it. Every insight should be supported by data.

5. Business Problems to Solve

Management has defined **five core problem areas** for the Analytics Team to investigate. Each problem represents a real operational challenge with measurable business impact.

Problem 1: Delivery Delays

Customer complaints regarding late deliveries have increased significantly, leading to poor ratings and reduced repeat orders.

Q1	Which restaurants are associated with the longest average delivery times?
Q2	Which delivery partners have the highest average time per order?
Q3	Which cities experience the most delivery delays?
Q4	Are delivery delays increasing or decreasing over time?

Goal: Reduce delivery times and restore customer satisfaction scores.

Problem 2: Order Cancellations

Cancellation rates have climbed over the past six months, creating revenue leakage and wasted operational resources.

Q1	Which restaurants receive the highest volume of cancellations?
Q2	Which cities have the highest cancellation rates by percentage?
Q3	Is there a correlation between delivery delays and cancellations?
Q4	On which days and at which times do cancellations peak?

Goal: Identify cancellation drivers and reduce revenue loss through targeted intervention.

Problem 3: Revenue Optimisation

Revenue growth is inconsistent across cities, restaurants, and time periods. Management wants to identify and replicate success patterns.

Q1	Which restaurants generate the highest total revenue?
Q2	Which cities contribute the most to overall company revenue?
Q3	Which payment methods are most popular and most valuable?
Q4	What are the monthly revenue trends over the past year?

Goal: Maximise profitability through data-driven resource allocation and marketing.

Problem 4: Customer Retention

Acquiring a new customer costs five times more than retaining an existing one. Identifying and re-engaging at-risk customers is a strategic priority.

Q1	Who are the top 10 most valuable customers by lifetime spending?
Q2	Which customers order most frequently, and what is their average order value?
Q3	Which customers have gone inactive in the last 90 days?
Q4	What behavioural patterns signal that a customer is about to churn?

Goal: Build targeted retention campaigns based on customer behaviour segments.

Problem 5: Delivery Partner Performance

Delivery partners are the face of the company. Poor partner performance directly damages customer experience and brand reputation.

Q1	Who are the fastest delivery partners based on average delivery time?
Q2	Which partners handle the highest volume of successful deliveries?
Q3	Which partners are associated with the most cancellations or delays?
Q4	What is the average customer rating received per delivery partner?

Goal: Reward top performers, identify partners needing support, and improve SLA compliance.

6. Dataset & Table Structure

The project uses **seven CSV files** representing the complete operational data of the food delivery company. Each file maps to a table in the relational database.

customers.csv | 1,000 rows

Master record of all registered customers on the platform.

Column Name	Description
customer_id	Unique identifier for each customer
customer_name	Full name of the customer
age	Customer age in years
gender	Male / Female / Other
city	City where the customer is located
signup_date	Date the customer registered on the platform

restaurants.csv | 100 rows

Details of all restaurant partners listed on the platform.

Column Name	Description
restaurant_id	Unique identifier for each restaurant
restaurant_name	Name of the restaurant
cuisine_type	Category of food served
city	City where the restaurant operates
rating	Platform rating out of 5.0

orders.csv | 10,000 rows

Central fact table linking customers, restaurants, and delivery partners.

Column Name	Description
order_id	Unique identifier for each order
customer_id	Links to customers table
restaurant_id	Links to restaurants table
partner_id	Links to delivery_partners table

order_time	Timestamp when the order was placed
delivered_time	Timestamp when order was delivered (blank if not delivered)
order_amount	Value of items ordered in INR
delivery_fee	Delivery charge applied to the order
status	Delivered / Cancelled / Pending

delivery_partners.csv | 200 rows

Information about all registered delivery executives.

Column Name	Description
partner_id	Unique identifier for each delivery partner
partner_name	Full name of the delivery partner
vehicle_type	Bike / Scooter / Bicycle / Electric Bike
joining_date	Date the partner joined the platform

payments.csv | 10,000 rows

Payment transaction record for each order placed.

Column Name	Description
payment_id	Unique identifier for each payment
order_id	Links to orders table
payment_mode	UPI / Credit Card / Debit Card / Cash on Delivery / Wallet
amount	Total amount paid including delivery fee in INR

ratings.csv | 7,000 rows

Post-delivery customer feedback for a subset of delivered orders.

Column Name	Description
rating_id	Unique identifier for each rating record
order_id	Links to orders table
customer_rating	Rating given by the customer out of 5.0
feedback	Free-text comment written by the customer

customer_behavior.csv | 1,000 rows

Derived analytics table with aggregated customer behaviour metrics — used for Data Science projects.

Column Name	Description
customer_id	Links to customers table
total_orders	Total number of successfully delivered orders
total_spending	Cumulative spend across all orders in INR
avg_order_value	Average spend per delivered order
last_order_days	Number of days since last order as of January 2025
churn_flag	1 = Churned (inactive over 90 days or zero orders), 0 = Active

7. Expected Deliverables

Students must submit **six deliverables** to successfully complete the project. Each deliverable builds on the previous one.

D1 | Database Design (ER Diagram)

Description	Create an Entity-Relationship Diagram showing all six entities, their attributes, primary keys, foreign keys, and the cardinality of each relationship (one-to-many, many-to-one). The diagram should be drawn using a tool such as MySQL Workbench, dbdiagram.io, or draw.io.
SQL Topics	ER Modelling, Normalisation, Relationships

D2 | Table Creation Scripts (DDL)

Description	Write complete SQL CREATE TABLE scripts for all tables. Each script must include appropriate data types, NOT NULL constraints, PRIMARY KEY definitions, FOREIGN KEY references, and any CHECK constraints for data validation (e.g., rating must be between 1.0 and 5.0).
SQL Topics	CREATE TABLE, Constraints, Data Types, Primary Key, Foreign Key

D3 | Data Import & Validation

Description	Import all seven CSV files into your MySQL database. After each import, run a COUNT(*) query to verify that the correct number of records was loaded. Submit screenshots or a written log confirming successful data import.
SQL Topics	LOAD DATA, INSERT, COUNT(), Data Validation

D4 | Data Cleaning Report

Description	Perform a full data quality audit on all tables. Identify and resolve: NULL values in critical columns, duplicate order or payment records, invalid ratings outside the 1.0–5.0 range, inconsistent city name spellings, and orders with missing delivery timestamps. Document every issue found and every SQL statement used to fix it.
SQL Topics	IS NULL, TRIM(), REPLACE(), COALESCE(), DISTINCT, UPDATE, DELETE

D5 | SQL Analysis Queries (Minimum 40)

Description	Write and execute a minimum of 40 SQL queries answering the business questions across all five problem areas. Queries must span three difficulty levels: Basic (SELECT, WHERE, GROUP BY), Intermediate (JOINs, Subqueries, CASE statements), and Advanced (CTEs, Window Functions, Stored Procedures). Each query must be accompanied by a brief explanation of what it answers and a summary of the key finding.
SQL Topics	Full SQL syllabus — see Section 8

D6 | Business Insights Report

Description	Prepare a written analytical report (minimum 5 pages) containing: a summary of key findings for each of the five problem areas, the root causes identified through data analysis, specific and measurable recommendations for management, and the estimated business impact of implementing each recommendation. The report should be written in professional language suitable for a non-technical executive audience.
SQL Topics	Data Storytelling, Business Communication, Analytical Writing

8. SQL Concepts Covered

This project covers the **complete SQL syllabus** across three difficulty levels — from basic retrieval to advanced analytical functions. Each concept is applied directly to real business questions in the project.

BEGINNER LEVEL

Basic SQL	
Commands: SELECT | DISTINCT | WHERE | ORDER BY | LIMIT	**Used For:** Retrieve and filter orders, list unique cities, find top-rated restaurants.
Aggregation & Grouping	
Commands: COUNT() | SUM() | AVG() | MIN() | MAX() | GROUP BY | HAVING	**Used For:** Calculate total revenue per city, average delivery time, cancellation counts.
Data Manipulation (DML)	
Commands: INSERT | UPDATE | DELETE | TRUNCATE	**Used For:** Fix NULL values, remove duplicates, update incorrect city names.

INTERMEDIATE LEVEL

Joins	
Commands: INNER JOIN | LEFT JOIN | RIGHT JOIN | SELF JOIN	**Used For:** Combine orders with customer names, link payments to orders, match ratings to restaurants.
Subqueries	
Commands: Scalar Subqueries | Correlated Subqueries | Nested Queries | IN | EXISTS	**Used For:** Find customers spending above average, detect restaurants with no ratings.
Conditional Logic	
Commands: CASE WHEN ... THEN ... ELSE ... END | IF()	**Used For:** Classify delivery speed into Fast / Normal / Slow, segment customers by spend tier.
Data Cleaning Functions	
Commands: TRIM() | REPLACE() | COALESCE() | IFNULL() | NULLIF() | STR_TO_DATE()	**Used For:** Standardise city names, handle NULL delivery timestamps, correct data formats.

ADVANCED LEVEL

Common Table Expressions (CTEs)	
Commands: WITH cte AS (...) | Chained CTEs | Recursive CTEs	**Used For:** Build multi-step revenue analysis queries and customer cohort tracking reports.
Window Functions	

Commands: ROW_NUMBER() | RANK() | DENSE_RANK() | LEAD() | LAG() | PARTITION BY	**Used For:** Rank delivery partners by speed, calculate month-over-month revenue growth.
Database Objects	
Commands: CREATE VIEW | CREATE PROCEDURE | CALL procedure() | DROP VIEW	**Used For:** Create reusable revenue views and automated monthly report stored procedures.

9. Data Science Connection

This project is designed as a **bridge between SQL and Data Science**. The same dataset you analyse with SQL can later be used in Python with Pandas, Scikit-learn, and visualisation libraries for end-to-end machine learning projects.

Customer Churn Prediction	Type: **Binary Classification**
Target: churn_flag (1 = Churned)	**Algorithms:** Logistic Regression, Random Forest, XGBoost
Key Features: total_orders, avg_order_value, last_order_days, city, signup_date	

Delivery Time Prediction	Type: **Regression**
Target: Delivery time in minutes	**Algorithms:** Linear Regression, Gradient Boosting, Neural Networks
Key Features: restaurant_id, partner_id, city, order_time, order_amount	

Order Cancellation Prediction	Type: **Binary Classification**
Target: status = Cancelled (1/0)	**Algorithms:** Decision Tree, SVM, Ensemble Methods
Key Features: restaurant_id, city, order_amount, day_of_week, hour_of_day	

Customer Segmentation	Type: **Unsupervised Clustering**
Target: No label — natural clusters	**Algorithms:** K-Means, DBSCAN, Hierarchical Clustering
Key Features: total_orders, total_spending, avg_order_value, last_order_days	

10. Success Criteria

A student successfully completes this project when they can demonstrate the following competencies. These mirror the skills assessed in a real-world Data Analyst job interview.

#	Item

1	Design a normalised relational database with correct primary and foreign key constraints.
2	Create and manage multiple related tables using DDL commands.
3	Import large datasets from CSV files and validate data integrity after loading.
4	Write complex SQL queries using joins, subqueries, CTEs, and window functions.
5	Clean raw data by identifying and resolving NULLs, duplicates, and formatting errors.
6	Generate at least 40 analytical SQL queries answering real business questions.
7	Interpret query results and translate them into clear business insights.
8	Provide specific, measurable, and data-supported recommendations to management.
9	Prepare a clean, feature-engineered dataset ready for Data Science and ML projects.
10	Present findings in a professional report suitable for a non-technical audience.

This project simulates the day-to-day responsibilities of a real-world Data Analyst. Every query you write, every insight you generate, and every recommendation you make reflects the kind of work that business intelligence teams do in leading companies. Approach each deliverable with the rigour and professionalism expected in industry.